

AARADHYA PANDEY

Postdoctoral Fellow in Statistics, Harvard University

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Employment and Education

Harvard University (advised by Morgane Austern)

Postdoctoral fellow in the Statistics department

Harvard

Sep. 2026 - present

Princeton University (advised by Sanjeev Kulkarni and Arian Maleki)

PhD in ORFE with Gordon Wu fellowship: ‘awarded to the most outstanding incoming doctoral students in engineering’

Princeton

Sep. 2021 - Aug. 2026

Indian Institute of Science (IISc)

Bachelor of Science in mathematics with CGPA 9.6/10 : discipline rank 1, was awarded the institute gold medal.

Bangalore

Sep. 2017 - Jun. 2021

Research Interests

My research interests are at the interface of probability theory, mathematical statistics, machine learning, and information theory with applications in (quantum) differential privacy, machine unlearning and spin glasses.

Publications and Preprints

- [1] **Statistical unlearning of distributions: a hypothesis testing approach.** Aaradhya Pandey, Sanjeev Kulkarni. arXiv:2605.16645
- [2] **Infinitely divisible privacy and beyond I: Resolution of the $s^2 = 2k$ conjecture.** Aaradhya Pandey, Arian Maleki, Sanjeev Kulkarni. arXiv:2512.00734
- [3] **Gaussian certified unlearning in high dimensions: a hypothesis testing approach.** Aaradhya Pandey, Arnab Auddy, Haolin Zou, Arian Maleki, Sanjeev Kulkarni. arXiv:2510.13094 (accepted to ICLR 2026 for oral presentation)
- [4] **Exact recovery in Gaussian weighted stochastic block model and planted dense subgraphs: statistical and algorithmic thresholds.** Aaradhya Pandey, Sanjeev Kulkarni. arXiv:2402.12515 (submitted)
- [5] **Community detection in the hypergraph stochastic block model and reconstruction on hypertrees.** Yuzhou Gu, Aaradhya Pandey. *Proceedings of the 37th Conference on Learning Theory*, PMLR 247:2166–2203. PDF | Proceedings

Projects in Preparation

- [1] **Quantum f -differential privacy: a hypothesis testing approach.** Aaradhya Pandey, Morgane Austern.
- [2] **Quantum infinitely divisible states: a genuinely quantum phenomenon.** Aaradhya Pandey, Morgane Austern.
- [3] **Multivariate version of the Ghirlanda–Guerra identities and the matrix of spin correlations.** Aaradhya Pandey.

Invited Talks and Presentations

Summer 2026	Presented my work on privacy at AIM Caltech, ICERM Brown, One World MINDS	USA
Early 2026	Presented my work on privacy at Johns Hopkins, Google NYC, Columbia, Harvard	USA
March 2026	Presented my work on privacy at Institute of Science and Technology, Austria	Austria
Jan 2026	Presented my work on differential privacy at STCS TIFR, CSA IISc	India
May 2025	Gave a tutorial on Information theory and Differential privacy at TIFR CAM	Bangalore
May 2025	Presented my work on the Gaussian certified unlearning paper at IISc	Bangalore
Jan 2025	Presented my work on the Correlation matrix of spin glasses at IISc and TIFR CAM	Bangalore
Oct 2024	Presented my work on Stochastic block model papers at Michigan State University	Michigan
Feb 2024	Our joint work on Stochastic block model paper was presented at the Institute of Advanced Study	Princeton

Academic Service

Reviewing I	Information and Inference, Foundations of Computational Mathematics.	<i>Princeton</i>
Reviewing II	IEEE Transactions on Information theory, NeurIPS 2026, ICLR 2027	<i>Princeton</i>
Teaching I	Undergraduate probability theory, statistics, game theory, networks, signals, and systems classes	<i>Princeton</i>
Teaching II	A 24-hour course for master's students covering probability, martingales, and Markov chains	<i>Princeton</i>
Organizing	Spanning over several semesters a student reading group in High dimensional Probability and Statistics	<i>Princeton</i>

Selected Coursework

Credited	Topics classes in High-dimensional probability (ORF 550, Grade A), Statistical machine learning (ORF 570, Grade A), Probabilistic methods (MAT 577, Grade A), Nonlinear models in econometrics (ECO 519, Grade A)	<i>Princeton</i>
Audited	Classes in Phase transitions (PHY 535), Combinatorial optimization (MAT 572), Computational methods (MAT 586), Discrete probability (MAT 589), Condensed matter (MAT 595), Statistical mechanics (MAT 597)	<i>Princeton</i>

Fellowships and Achievements

2021 – 2026	Gordon Wu fellow : Awarded to the most outstanding incoming doctoral students in engineering	<i>Princeton</i>
NET 2020	Cleared with an all-India rank 1 in the entrance exam for mathematics PhD programs in India	<i>India</i>
2017 – 2021	KVPY fellow : Prestigious fellowship program for Indian undergraduate students interested in science	<i>Bangalore</i>
IIT JEE 2017	Cleared with an all-India rank of 305 (1 million participants) for undergraduate admission at IITs	<i>India</i>
Summer 2020	DAAD WISE : Prestigious fellowship for a funded summer project in Germany	<i>Bonn</i>

References

- **Morgane Austern**, Assistant Professor of Statistics, Harvard University ✉ maustern@fas.harvard.edu
- **Yuzhou Gu**, Member of Technical Staff, Open AI ✉ sevenkplus.g@gmail.com
- **Arnab Auddy**, Assistant Professor of Statistics, Ohio State University ✉ auddy.1@osu.edu
- **Arian Maleki**, Professor of Statistics, Columbia University ✉ arian@stat.columbia.edu
- **Sanjeev Kulkarni**, William R. Kenan Jr. Professor of ECE and ORFE, Princeton University ✉ kulkarni@princeton.edu